Goal: Use the tools used to find maximums and minimums in applications.

Main Idea: Given a situation, find the "optimal" value (usually a _____ or _____).

Example 1. A farmer is building a rectangular fence and has $400'$ of fencing to work with. What should the dimensions of the fence be to make the area as large as possible?

The function we want to optimize is called the _____ function. What is it for this example?

In order to reduce the number of variables down to 1 and/or guarantee a max/min, we need more conditions called _____. Are there any in this example?

We can now finish the problem (by finding the absolute max).

Example 2. Suppose that two numbers x and y add up to $S > 0$. When is their product at a maximum?

Example 3. A $36'' \times 72''$ piece of cardboard is cut from each corner and the sides are folded up to create a box with an open top. How far from the edge should the cut be to maximize the volume?

Example 4. You are making a rectangular garden and you have $50'$ of fencing to work with. To make a bigger garden, you build it right next to your house. How large can the garden be?

Example 5. A poster board has an area of $180\,\text{cm}^2$ and has margins as shown in the diagram below. What is the largest printable area?

